A Collaborative Submission by IEEE Student Branch TBS & IEEE Student Branch INSAT

LUMIO

Where Every Student Is Seen,
Every Struggle Is Understood
Every Parent Is Informed…

Lumio brings together AI engineering, clinical expertise, business strategy, and UX design to deliver a solution that is technically robust, ethically grounded, and human-centered. A currently enrolled medical student serves as our clinical advisor, ensuring every feature of the platform adheres to evidence-based methodologies and responsible practices in neurodevelopmental support.

Table Of Contents

1. Problem Statement: ADHD and Its Impact on Learning ……01

1.1 Overview and Prevalence of ADHD
1.2 Academic and Cognitive Impact
1.3 Challenges and Support Gaps

2. Solution Overview ……...02

2.1 What is LUMIO?
2.2 How does it work?
2.3 Data Architecture & Privacy

3. Expected Impact ……03

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1. Problem Statement: ADHD and Its Impact on Learning

1.1 Overview and Prevalence of ADHD

Attention-Deficit/Hyperactivity Disorder (ADHD) is a chronic **neurodevelopmental condition** characterized by inattention, hyperactivity, and impulsivity. It is one of the most prevalent learning barriers in educational settings worldwide. ADHD affects **7.6%** of children aged **3–12** and **5.6%** of adolescents aged **12–18** globally (Frontiers in Psychology, 2025).

Tunisia is no exception. A study conducted in Monastir (2022) found that **18.1%** of Tunisian high school students exhibit symptoms of ADHD, with strong links to poor parental relationships, psychiatric history, and academic underperformance. Similarly, a school-based study in Sfax (2021) reported a prevalence of **9.94%** among primary school children. Notably, **72.54%** of these children presented at least one comorbid disorder, including anxiety (**31.37%**) and learning disabilities (**23.52%**). This highlights the complexity of ADHD and the need for multidimensional support systems.

1.2 Academic and Cognitive Impact

ADHD significantly affects cognitive functions, particularly working memory and attention control. As a result, **33.2%** of affected students fail to graduate on time, and only **0.06%** go on to earn a graduate degree (Springer, 2022; PMC, 2024). These outcomes reflect the long-term academic impact of insufficient support.

1.3 Challenges and Support Gaps

Teachers also face critical challenges. Nearly one-third have **limited knowledge** of ADHD and often struggle to distinguish between a student with ADHD and one who is simply disengaged. Without proper data or tools, early identification becomes difficult, and only **32%** of students receive any form of behavioral classroom support (ADHDAdvisor, 2024).

Parents are equally affected by this gap. In most cases, they receive feedback only during report card periods, with little to no visibility into their child's daily academic and behavioral challenges.

The issue, therefore, is not purely medical. Rather, it lies in the **lack of accessible and coordinated tools**. Students, teachers, and parents currently lack the means to act early, share insights, and respond effectively. This gap is fundamentally technological, limiting the ability to provide timely and collaborative support.

2. Solution Overview

2.1 What is LUMIO?

Lumio is an **AI-powered learning companion platform** that connects students, teachers, and parents within a single intelligent ecosystem, each with their own tailored interface, all synchronized in real time. By tracking focus through the device camera, learning from behavioral patterns over time, and translating that understanding into personalized guidance, Lumio delivers a secure, privacy-first experience across both **web and mobile**.

2.2 How does it work?

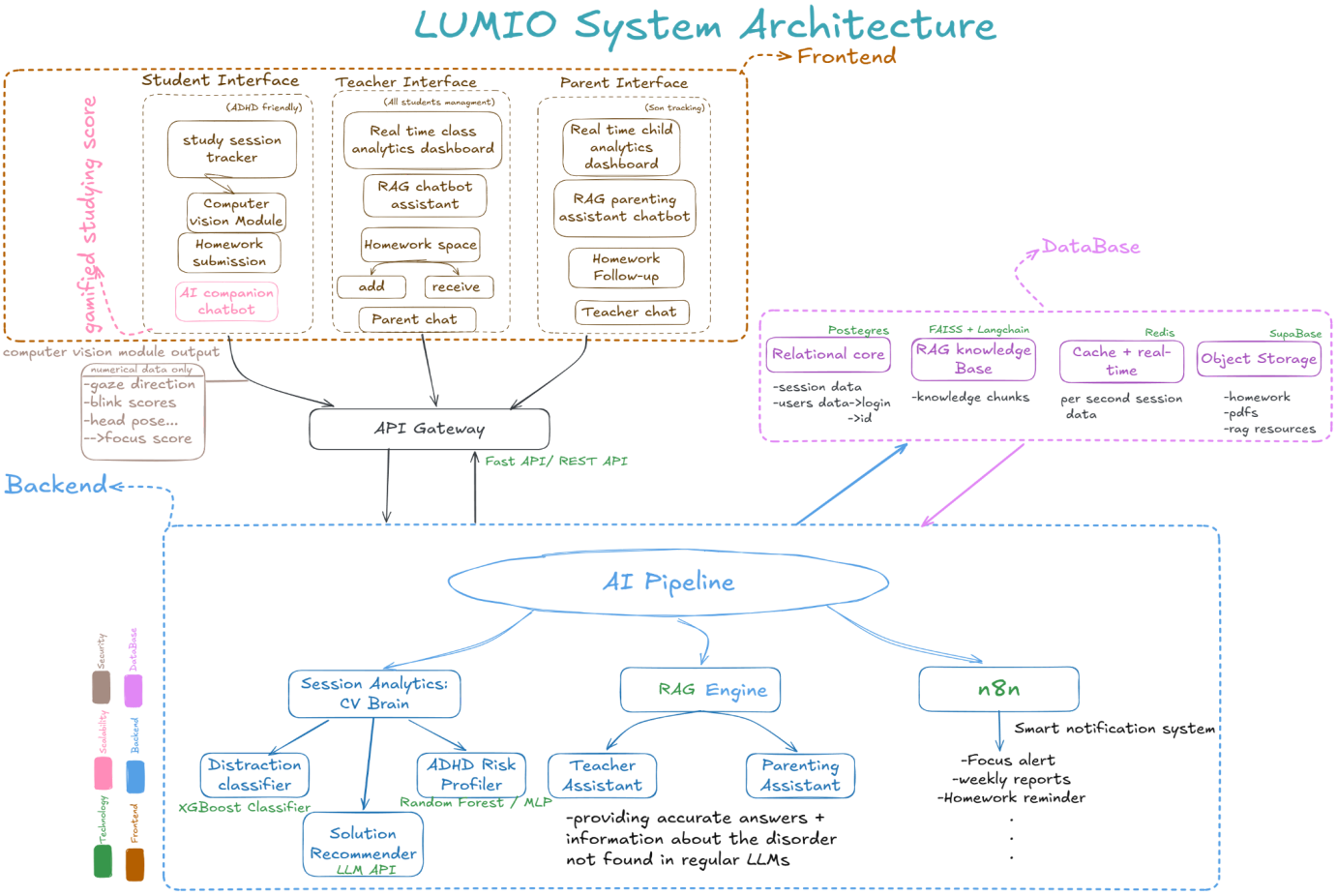
The experience begins the moment a student opens a session. The **Computer Vision module**, powered by **MediaPipe**, silently analyses gaze, blink rate, and head pose through the device camera — processing and discarding every frame locally so that only a compact **numerical focus score** ever reaches the server.

Those scores feed the **Session Analytics service**, where an **XGBoost classifier** labels each distraction event by cause, while **Random Forest and MLP models** build a longitudinal attention profile from 30 days of history and the form submitted by parents. A **deterministic rule engine** then interprets both outputs and maps them to a situation archetype — ensuring that consequential recommendations remain auditable and grounded in explicit logic rather than opaque prediction.

From there, the **RAG engine** — built with **LangChain** and **FAISS** — acts as the shared knowledge layer powering all three AI assistants. It retrieves disorder-specific passages from a curated

research library and injects them as context into the **teacher assistant**, the **parent assistant**, and the **student companion**, ensuring every response is evidence-grounded and tailored to its audience.

Finally, all resulting actions, alerts, reports, reminders, and notifications are orchestrated end-to-end by **n8n**, keeping every workflow transparent and entirely outside the core codebase.



TECHNOLOGY STACK

The following reflects our **preliminary implementation choices** and remains open to revision as the project matures.

React + React Native	***FastAPI · Python***	***LangChain · FAISS***	***XGBoost · MLP***	***MediaPipe WASM***	***n8n Automation***
PostgreSQL	***Redis***	***Supabase Storage***	***Claude / OpenAI API***	***Socket.io · WSS***	***Docker · Railway***

2.3 Data Architecture & Privacy

Lumio is built on four purpose-built stores: **PostgreSQL** for relational data, **Redis** for live state and job queuing, **FAISS** for the RAG vector index, and **Supabase Storage** for files and reports. No raw video is ever stored — the CV module processes every frame locally and transmits nothing but numbers. Where the attention risk profiler is concerned, its output is always a **risk score, not a diagnosis**, with professional evaluation recommended whenever that score crosses a defined threshold.

3. Expected Impact

Lumio serves all learners, especially those with cognitive or neurodiversity challenges such as **ADHD.**. Adaptive exercises and personalized content enhance focus, memory retention, and engagement, with AI-based tutoring and gamified learning resulting in up to a **25% increase** in engagement. This supports **SDG 4 – Quality Education, Target 4.5**, by promoting equitable learning opportunities for all students.

Our platform automatically flags patterns such as missed assignments, attention shifts, or spelling inconsistencies, enabling timely intervention. Predictive analytics studies report detection rates of **70–85%** for at-risk students, supporting our **80% early-alert goal** and aligning with **SDG 3 – Good Health and Well-being, Target 3.4** for early mental health support.

Inclusive learning is central to Lumio's design. By prioritizing accessibility and equity, the platform embodies **SDG 10 – Reduced Inequalities, Target 10.2**, ensuring that all learners have equal opportunities to progress and belong.

Interactive dashboards empower teachers and caregivers with engagement trends, risk alerts, and recommended actions, enabling collaboration and better student support at school and home. Studies show such tools can boost teacher confidence by **~30%**, aligning with **SDG 4 and SDG 3.**

Ethical and scalable design underpins the platform. Privacy-preserving AI processes sensitive data locally, storing only anonymized engagement metrics. This ensures compliance with **ethical** standards and builds trust among students, families, and institutions while enabling broad deployment.